

DDA3020 Machine Learning

Lecture 06 Logistic Regression

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Outline

- 1 Review of last week
- 2 Classification and representation
- 3 Logistic regression
- 4 Regularized logistic regression
- 5 Probabilistic perspective of logistic regression
- 6 Summary: linear regression vs. logistic regression

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Linear regression: deterministic perspective

- **Linear hypothesis function:** $f_{\mathbf{w}, w_0}(\mathbf{x}) = \mathbf{x}^\top \mathbf{w} + w_0$, or, simply $f_{\mathbf{w}}(\mathbf{x}) = \mathbf{x}^\top \mathbf{w}$ by concatenating w_0 and $\mathbf{w}$ together and augmenting $\mathbf{x}$ to $[1; \mathbf{x}]$
- **Linear regression** by minimizing the residual sum of squares (RSS):

$$\mathbf{w}^* = \arg \min_{\mathbf{w}} J(\mathbf{w}), \text{ where } J(\mathbf{w}) = \frac{1}{2} \sum_{i=1}^m (\mathbf{x}_i^\top \mathbf{w} - y_i)^2 = \frac{1}{2} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|^2$$

- **Two solutions:**
 - Closed-form solution: $\mathbf{w}^* = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$
 - Gradient descent: $\mathbf{w} \leftarrow \mathbf{w} - \alpha \mathbf{X}^\top (\mathbf{X}\mathbf{w} - \mathbf{y})$, for multiple iterations until convergence

Linear regression: probabilistic perspective

- We assume that: $y = \mathbf{w}^\top \mathbf{x} + e$, where $e \sim \mathcal{N}(0, \sigma^2)$ is called **observation noise** or **residual error**
- y is also a random variable, and its conditional probability is

$$p(y|\mathbf{x}, \mathbf{w}) = \mathcal{N}(\mathbf{w}^\top \mathbf{x}, \sigma^2)$$

- **Maximum log-likelihood estimation:**

$$\mathbf{w}_{MLE} = \arg \max_{\mathbf{w}} \log \mathcal{L}(\mathbf{w}; D) = \arg \max_{\mathbf{w}} \log \left(\prod_i^m p(y_i | \mathbf{x}_i, \mathbf{w}) \right) \quad (1)$$

$$= \arg \max_{\mathbf{w}} \sum_i^m \log p(y_i | \mathbf{x}_i, \mathbf{w}) = \arg \max_{\mathbf{w}} \sum_i^m \log \mathcal{N}(\mathbf{w}^\top \mathbf{x}_i, \sigma^2) \quad (2)$$

$$= \arg \max_{\mathbf{w}} -\log(\sigma^m (2\pi)^{\frac{m}{2}}) - \frac{1}{2\sigma^2} \sum_i^m (y_i - \mathbf{w}^\top \mathbf{x}_i)^2 \quad (3)$$

$$= \arg \min_{\mathbf{w}} \frac{1}{2} \sum_i^m (y_i - \mathbf{w}^\top \mathbf{x}_i)^2, \quad (4)$$

Variants of linear regression

- **Ridge regression** to avoid over-fitting, through MAP estimation:

$$\mathbf{w}_{MAP} = \arg \max_{\mathbf{w}} \sum_{i=1}^m \log p(y_i | \mathbf{x}_i, \mathbf{w}) + \log p(\mathbf{w}) \quad (5)$$

$$= \arg \max_{\mathbf{w}} \sum_{i=1}^m \log \mathcal{N}(\mathbf{w}^\top \mathbf{x}_i, \sigma^2) + \mathcal{N}(\mathbf{w} | \mathbf{0}, \tau^2 \mathbf{I}) \quad (6)$$

$$\equiv \arg \min_{\mathbf{w}} \sum_{i=1}^m (\mathbf{w}^\top \mathbf{x}_i - y_i)^2 + \lambda \|\mathbf{w}\|_2^2. \quad (7)$$

- **Polynomial regression:** linear model with **basis expansion** $\phi(\mathbf{x})$

$$\begin{aligned} f_{\mathbf{w}}(\mathbf{x}) &= w_0 + \sum_{i=1}^d w_i x_i + \sum_{i=1}^d \sum_{j=1}^d w_{ij} x_i x_j + \sum_{i=1}^d \sum_{j=1}^d \sum_{k=1}^d w_{ijk} x_i x_j x_k + \dots \\ &= \phi(\mathbf{x})^\top \mathbf{w}, \end{aligned} \quad (8)$$

$$\phi(\mathbf{x}) = [1, x_1, \dots, x_d, \dots, x_i x_j, \dots, x_i x_j x_k, \dots]^\top,$$

$$\mathbf{w} = [w_0, w_1, \dots, w_d, \dots, w_{ij}, \dots, w_{ijk}, \dots]^\top.$$

Variants of linear regression

- **Lasso regression** to obtain sparse model,

$$\mathbf{w}_{MAP} = \arg \max_{\mathbf{w}} \sum_i^m \log \mathcal{N}(\mathbf{w}^\top \mathbf{x}_i, \sigma^2) + \text{Lap}(\mathbf{w} | \mathbf{0}, b) \quad (9)$$

$$= \arg \min_{\mathbf{w}} \sum_{i=1}^m (\mathbf{w}^\top \mathbf{x}_i - y_i)^2 + \lambda \|\mathbf{w}\|_1. \quad (10)$$

- **Robust regression** for data with outliers:

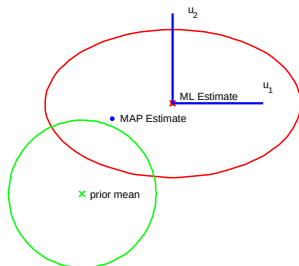
$$\mathbf{w}_{MLE} = \arg \min_{\mathbf{w}} \sum_{i=1}^m |\mathbf{w}^\top \mathbf{x}_i - y_i| \quad (11)$$

Summary of different linear regressions

Note that the uniform distribution will not change the mode of the likelihood.

Thus, MAP estimation with a uniform prior corresponds to MLE.

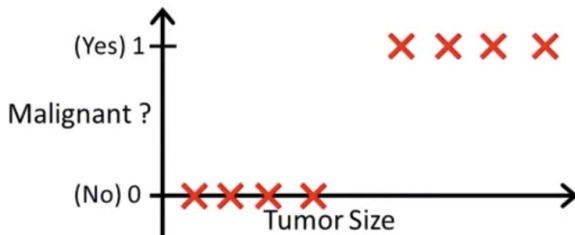
$p(y \mathbf{x}, \mathbf{w})$	$p(\mathbf{w})$	regression method
Gaussian	Uniform	Least squares
Gaussian	Gaussian	Ridge regression
Gaussian	Laplace	Lasso regression
Laplace	Uniform	Robust regression
Student	Uniform	Robust regression



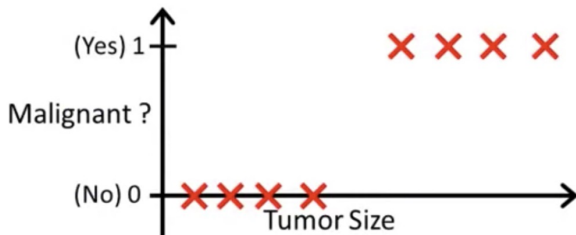
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Classification

- Classification: classifying input data into discrete states
 - Email filtering: spam / not spam?
 - Weather forecast: sunny / not sunny?
 - Tumor: malignant / benign?
- The label $y \in \{0, 1\}$:
 - $y = 0$: negative class, *e.g.*, not spam, not sunny, benign
 - $y = 1$: positive class, *e.g.*, spam, sunny, malignant

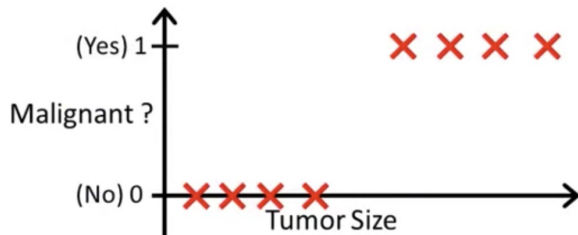


Threshold classifier with linear regression



- We assume a linear hypothesis function $f_{\mathbf{w}}(\mathbf{x}) = \mathbf{x}^{\top} \mathbf{w}$
- A simple threshold classifier with this hypothesis function is
 - If $f_{\mathbf{w}}(\mathbf{x}) > 0.5$, then $y = 1$, *i.e.*, malignant tumor
 - If $f_{\mathbf{w}}(\mathbf{x}) < 0.5$, then $y = 0$, *i.e.*, benign tumor

Threshold classifier with linear regression



- It seems that the simple threshold classifier with linear regression works well on this classification task
- However, if there is a positive sample with a very large tumor size ([plot above](#)), what will happen?
- The hypothesis function will be **significantly changed**, causing some positive samples to be misclassified as negative (not malignant). How to handle it? [Adjusting the threshold value](#), or adopting [robust linear regression](#).

Threshold classifier with linear regression

- But there is still something **wired**.
- Our goal is to predict $y \in \{0, 1\}$, but the prediction could be $f_{\mathbf{w}}(\mathbf{x}) > 1$ or $f_{\mathbf{w}}(\mathbf{x}) < 0$, which does not serve our purpose.
- A **desired hypothesis function** for this task should be $f_{\mathbf{w}}(\mathbf{x}) \in [0, 1]$.

Threshold classifier with linear regression

Exercise: Which statements are true?

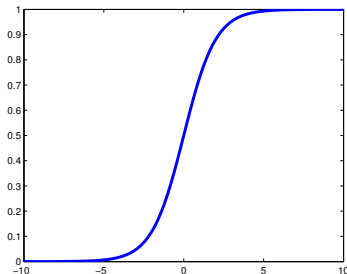
- If linear regression doesn't work well like the above example, feature scaling may help
- If the training set satisfies that all $y_i \in [0, 1]$ for all points $(\mathbf{x}_i, y_i)$, then the linear hypothesis function $f_{\mathbf{w}}(\mathbf{x}) \in [0, 1]$ for all values of $\mathbf{x}_i$
- None of the above is correct

Hypothesis representation

- A desired hypothesis function for this task should be $f_{\mathbf{w}}(\mathbf{x}) \in [0, 1]$
- To this end, we introduce a novel function, as follows:

$$f_{\mathbf{w}}(\mathbf{x}) = g(\mathbf{w}^\top \mathbf{x}) \in [0, 1], \quad g(z) = \frac{1}{1 + \exp(-z)},$$

where $g(\cdot)$ is called **sigmoid function** or **logistic function** (shown below)



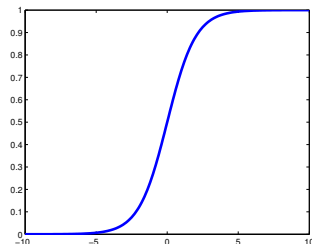
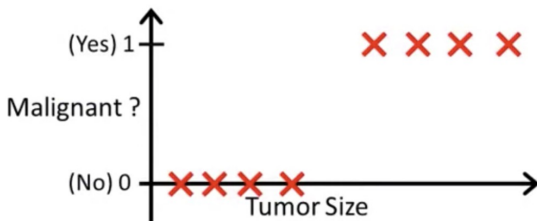
* Note that $\mathbf{w} = [w_0, w_1, \dots, w_d]^\top$ and $\mathbf{x} = [1, x_1, \dots, x_d]^\top$.

Hypothesis representation

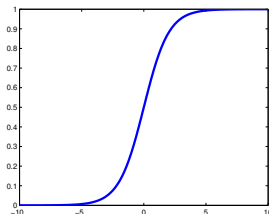
- **Interpretation of sigmoid/logistic function**

- $f_{\mathbf{w}}(\mathbf{x})$ = estimated probability that $y = 1$ of input $\mathbf{x}$.
- For example (plot below), if $f_{\mathbf{w}}(\mathbf{x}) = 0.8$, then it means that a patient with tumor size $\mathbf{x}$ has 80% chance of tumor being malignant. In this task, a larger tumor size has a larger chance/probability of being a malignant tumor.
- Thus, we can say that

$$f_{\mathbf{w}}(\mathbf{x}) = P(y = 1 | \mathbf{x}; \mathbf{w}).$$



Decision boundary

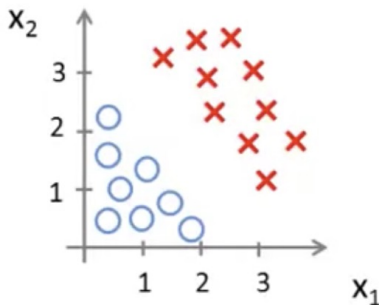


- In logistic regression, we have

$$f_{\mathbf{w}}(\mathbf{x}) = g(\mathbf{w}^T \mathbf{x}) = P(y = 1 | \mathbf{x}; \mathbf{w}) \in [0, 1], \quad g(z) = \frac{1}{1 + \exp(-z)}.$$

- Suppose that if $f_{\mathbf{w}}(\mathbf{x}) \geq 0.5$, then we predict $y = 1$; if $f_{\mathbf{w}}(\mathbf{x}) < 0.5$, then we predict $y = 0$
- Correspondingly, if $\mathbf{w}^T \mathbf{x} \geq 0$, we predict $y = 1$; if $\mathbf{w}^T \mathbf{x} < 0$, then we predict $y = 0$.
- It determines the **decision boundary**, which is the curve/hyper-plane corresponding to $f_{\mathbf{w}}(\mathbf{x}) = 0.5$, or $\mathbf{w}^T \mathbf{x} = 0$

Decision boundary



- $f_{\mathbf{w}}(\mathbf{x}) = g(w_0 + w_1x_1 + w_2x_2) = g(-3 + x_1 + x_2)$
- Predict $y = 1$ if $-3 + x_1 + x_2 \geq 0$ (plot above)

Decision boundary

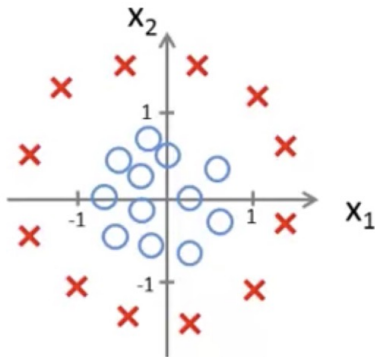


Figure: Non-linear decision boundary

- $f_{\mathbf{w}}(\mathbf{x}) = g(w_0 + w_1x_1 + w_2x_2 + w_3x_1^2 + w_4x_2^2) = g(-1 + x_1^2 + x_2^2)$
- Predict $y = 1$ if $-1 + x_1^2 + x_2^2 \geq 0$ (plot above)

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Cost function

- **Training set:** m training examples $\{(\mathbf{x}_i, y_i)\}_{i=1}^m$
- **Hypothesis function:** $f_{\mathbf{w}}(\mathbf{x}) = g(\mathbf{w}^\top \mathbf{x}) = \frac{1}{1 + \exp(-\mathbf{w}^\top \mathbf{x})}$
- **Cost function:**
 - **Linear regression:** $J(\mathbf{w}) = \frac{1}{2m} \sum_{i=1}^m (f_{\mathbf{w}}(\mathbf{x}_i) - y_i)^2 = \frac{1}{2m} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|^2$, which is called ℓ_2 loss or **residual sum of squares**
 - It is **convex** *w.r.t.* $\mathbf{w}$ for linear regression
 - **Logistic regression:** If we adopt the same cost function for logistic regression, we have

$$J(\mathbf{w}) = \frac{1}{2m} \sum_i^m (g(\mathbf{w}^\top \mathbf{x}_i) - y_i)^2.$$

However, it is **non-convex** *w.r.t.* $\mathbf{w}$.

Exercise 1: Prove the ℓ_2 loss is convex *w.r.t.* $\mathbf{w}$ for **linear regression**.

Exercise 2: Prove the ℓ_2 loss is non-convex *w.r.t.* $\mathbf{w}$ for **logistic regression**.

Cost function

Exercise 1: Prove the ℓ_2 loss is convex *w.r.t.* $\mathbf{w}$ for linear regression.

Exercise 2: Prove the ℓ_2 loss is non-convex *w.r.t.* $\mathbf{w}$ for logistic regression.

Cost function

- Cross-entropy:

$$H(p, q) = - \int_x p(x) \log(q(x)) dx \quad \text{or} \quad - \sum_x p(x) \log(q(x)),$$

where $p(x), q(x)$ are **probability density functions** (PDF) of x if x is a continuous random variable, **or, probability mass functions** if x is a discrete random variable

- We set

ground-truth posterior probability : $y(\mathbf{x}) = P(y = 1|\mathbf{x})$,

predicted posterior probability : $f_{\mathbf{w}}(\mathbf{x}) = P(y = 1|\mathbf{x}; \mathbf{w})$.

- Cross-entropy loss:

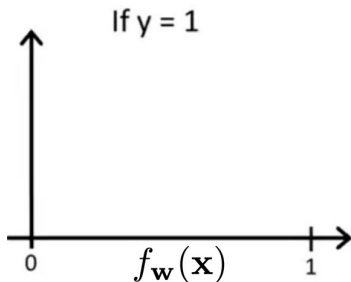
$$\begin{aligned} \text{cost}(y(\mathbf{x}), f_{\mathbf{w}}(\mathbf{x})) &= H(y(\mathbf{x}), f_{\mathbf{w}}(\mathbf{x})) \\ &= -P(y = 1|\mathbf{x}) \cdot \log P(y = 1|\mathbf{x}; \mathbf{w}) - P(y = 0|\mathbf{x}) \cdot \log P(y = 0|\mathbf{x}; \mathbf{w}) \\ &= \begin{cases} -\log(f_{\mathbf{w}}(\mathbf{x})), & \text{if } y(\mathbf{x}) = 1 \\ -\log(1 - f_{\mathbf{w}}(\mathbf{x})), & \text{if } y(\mathbf{x}) = 0 \end{cases} \end{aligned}$$

Cost function for logistic regression

- Cross-entropy loss:

$$\text{cost}(y(\mathbf{x}), f_{\mathbf{w}}(\mathbf{x})) = \begin{cases} -\log(f_{\mathbf{w}}(\mathbf{x})), & \text{if } y(\mathbf{x}) = 1 \\ -\log(1 - f_{\mathbf{w}}(\mathbf{x})), & \text{if } y(\mathbf{x}) = 0 \end{cases}$$

- For $y = 1$, if $f_{\mathbf{w}}(\mathbf{x}) = 1$, i.e., $P(y = 1|\mathbf{x}; \mathbf{w}) = 1$, then the prediction equals to the ground-truth label, the cost is 0.
- For $y = 1$, if $f_{\mathbf{w}}(\mathbf{x}) \rightarrow 0$, i.e., $P(y = 1|\mathbf{x}; \mathbf{w}) \rightarrow 0$, then it should be penalized with a very large cost. Here we have $\text{cost}(y(\mathbf{x}), f_{\mathbf{w}}(\mathbf{x})) \rightarrow \infty$.

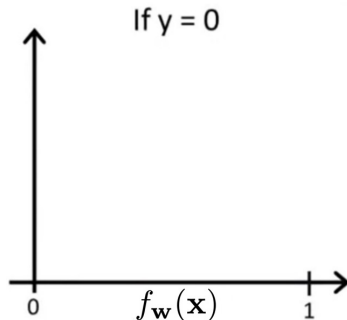


Cost function for logistic regression

- **Cross-entropy loss:**

$$\text{cost}(y(\mathbf{x}), f_{\mathbf{w}}(\mathbf{x})) = \begin{cases} -\log(f_{\mathbf{w}}(\mathbf{x})), & \text{if } y(\mathbf{x}) = 1 \\ -\log(1 - f_{\mathbf{w}}(\mathbf{x})), & \text{if } y(\mathbf{x}) = 0 \end{cases}$$

- For $y = 0$, if $f_{\mathbf{w}}(\mathbf{x}) = 0$, i.e., $P(y = 1|\mathbf{x}; \mathbf{w}) = 0$, then the prediction equals to the ground-truth label, the cost is 0
- For $y = 0$, if $f_{\mathbf{w}}(\mathbf{x}) \rightarrow 1$, i.e., $P(y = 1|\mathbf{x}; \mathbf{w}) \rightarrow 1$, then it should be penalized with a very large cost. Here we have $\text{cost}(y(\mathbf{x}), f_{\mathbf{w}}(\mathbf{x})) \rightarrow \infty$



Cost function for logistic regression

- **Cross-entropy loss:**

$$\text{cost}(y(\mathbf{x}), f_{\mathbf{w}}(\mathbf{x})) = \begin{cases} -\log(f_{\mathbf{w}}(\mathbf{x})), & \text{if } y(\mathbf{x}) = 1 \\ -\log(1 - f_{\mathbf{w}}(\mathbf{x})), & \text{if } y(\mathbf{x}) = 0 \end{cases}$$

Exercise: Which states are true?

- If $f_{\mathbf{w}}(\mathbf{x}) = y$, then $\text{cost}(y(\mathbf{x}), f_{\mathbf{w}}(\mathbf{x})) = 0$ for both $y = 0$ and $y = 1$
- If $y = 0$, then $\text{cost}(y(\mathbf{x}), f_{\mathbf{w}}(\mathbf{x})) \rightarrow \infty$ as $f_{\mathbf{w}}(\mathbf{x}) \rightarrow 1$
- If $y = 0$, then $\text{cost}(y(\mathbf{x}), f_{\mathbf{w}}(\mathbf{x})) \rightarrow \infty$ as $f_{\mathbf{w}}(\mathbf{x}) \rightarrow 0$
- Regardless whether $y = 0$ or $y = 1$, if $f_{\mathbf{w}}(\mathbf{x}) = 0.5$, then $\text{cost}(y(\mathbf{x}), f_{\mathbf{w}}(\mathbf{x})) > 0$

Cost function of logistic regression

- Cost function of logistic regression

$$J(\mathbf{w}) = \frac{1}{m} \sum_{i=1}^m \text{cost}(y_i, f_{\mathbf{w}}(\mathbf{x}_i)),$$
$$\text{cost}(y(\mathbf{x}), f_{\mathbf{w}}(\mathbf{x})) = \begin{cases} -\log(f_{\mathbf{w}}(\mathbf{x})), & \text{if } y(\mathbf{x}) = 1 \\ -\log(1 - f_{\mathbf{w}}(\mathbf{x})), & \text{if } y(\mathbf{x}) = 0 \end{cases}$$

- The above cost function can be simplified as follows

$$J(\mathbf{w}) = -\frac{1}{m} \sum_{i=1}^m [y_i \log(f_{\mathbf{w}}(\mathbf{x}_i)) + (1 - y_i) \log(1 - f_{\mathbf{w}}(\mathbf{x}_i))].$$

Exercise: Please prove that $J(\mathbf{w})$ is convex *w.r.t.* $\mathbf{w}$.

Gradient descent for logistic regression

- Learning $\mathbf{w}$ by minimize $J(\mathbf{w})$, *i.e.*,

$$\mathbf{w}^* = \arg \min_{\mathbf{w}} J(\mathbf{w}) = -\frac{1}{m} \sum_{i=1}^m [y_i \log(f_{\mathbf{w}}(\mathbf{x}_i)) + (1 - y_i) \log(1 - f_{\mathbf{w}}(\mathbf{x}_i))].$$

- **Gradient descent:** repeat the following update until convergence

$$\begin{aligned}\mathbf{w} &\leftarrow \mathbf{w} - \alpha \nabla_{\mathbf{w}} J(\mathbf{w}) \\ \nabla_{\mathbf{w}} J(\mathbf{w}) &= \frac{1}{m} \sum_{i=1}^m [f_{\mathbf{w}}(\mathbf{x}_i) - y_i] \mathbf{x}_i\end{aligned}$$

- **How to define convergence?** Calculating the changes of $J(\mathbf{w})$ or $\mathbf{w}$ in the last K steps, if the change is lower than a threshold than it can be seen as convergence. Remember that choosing a suitable learning rate α is important to achieve a good converged solution.

Gradient descent for logistic regression

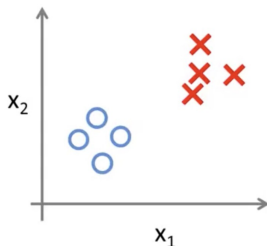
Exercise: Suppose you are running a logistic regression model, and you should observe the learning procedure to find a suitable learning rate α . Which of the following is reasonable to make sure α is set properly and that the gradient descent is running correctly?

- Plot $J(\mathbf{w}) = -\frac{1}{m} \sum_i^m (y_i - f_{\mathbf{w}}(\mathbf{x}_i))^2$ as a function of the number of iterations (*i.e.*, the horizontal axis is the iteration number) and make sure $J(\mathbf{w})$ is decreasing on every iteration.
- Plot $J(\mathbf{w}) = -\frac{1}{m} \sum_i^m [y_i \log(f_{\mathbf{w}}(\mathbf{x}_i)) + (1 - y_i) \log(1 - f_{\mathbf{w}}(\mathbf{x}_i))]$ as a function of the number of iterations (*i.e.*, the horizontal axis is the iteration number) and make sure $J(\mathbf{w})$ is decreasing on every iteration.
- Plot $J(\mathbf{w})$ as a function of $\mathbf{w}$ and make sure it is decreasing on every iteration.
- Plot $J(\mathbf{w})$ as a function of $\mathbf{w}$ and make sure it is convex.

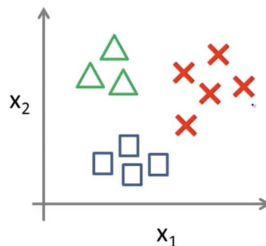
Multi-class classification

- **Binary classification:** in above examples and derivations, we only consider the **binary classification** problem, *i.e.*, $y \in \{0, 1\}$.
- **Multi-class/multi-category classification:** however, many practical problems involve with multi-category outputs, *i.e.*, $y \in \{1, \dots, C\}$:
 - **Weather forecast:** sunny, cloudy, rain, snow
 - **Email tagging:** work, friends, families, hobby

Binary classification:

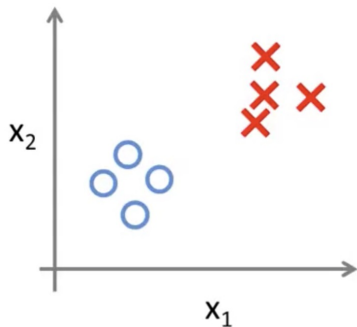


Multi-class classification:

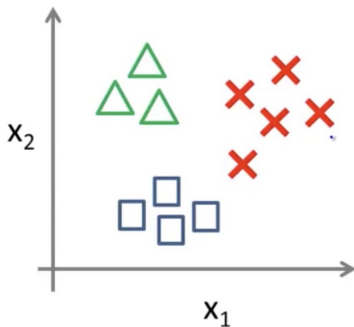


Multi-class classification

Binary classification:

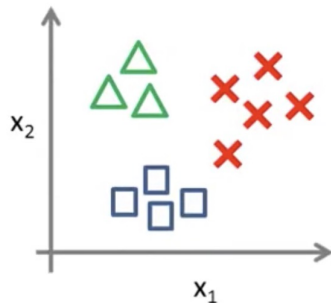


Multi-class classification:



Multi-class classification: one-vs-all

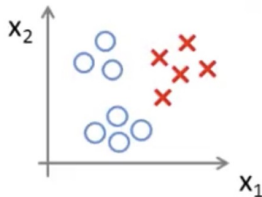
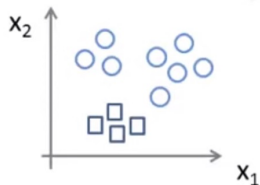
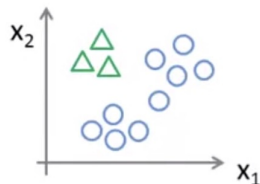
One-vs-all (one-vs-rest):



Class 1: 

Class 2: 

Class 3: 



Multi-class classification: one-vs-all

One-vs-all logistic regression:

- Train a binary logistic regression $f_{\mathbf{w}_j}(\cdot)$ for each class j , by setting all samples of other classes as negative class
- For a new testing sample $\mathbf{x}$, predict its class as $\arg \max_j f_{\mathbf{w}_j}(\mathbf{x})$.

Pros: Easy to implement

Cons: The training cost is too high and is difficult to scale to tasks with a large number of classes.

Multi-class classification: Softmax regression

- Softmax function:

$$f_{\mathbf{W}}^{(j)}(\mathbf{x}) = \frac{\exp(\mathbf{w}_j^\top \mathbf{x})}{\sum_{c=1}^C \exp(\mathbf{w}_c^\top \mathbf{x})} = P(y = j | \mathbf{x}; \mathbf{W}), \quad (12)$$

where $\mathbf{W} = [\mathbf{w}_1, \dots, \mathbf{w}_C] \in \mathbb{R}^{(d+1) \times C}$ with C being the number of classes and $\mathbf{x} = [1, x_1, \dots, x_d]^\top$ is the augmented feature vector. For simplicity, in the following we write $f_{\mathbf{W}}^{(j)}(\cdot)$ as $f_{\mathbf{w}_j}(\cdot)$.

- Cost function:

$$J(\mathbf{W}) = -\frac{1}{m} \sum_i^m \sum_j^C [\mathbb{I}(y_i = j) \log(f_{\mathbf{w}_j}(\mathbf{x}_i))], \quad (13)$$

where $\mathbb{I}(a) = 1$ if a is true, otherwise $\mathbb{I}(a) = 0$.

Multi-class classification: Softmax regression

- It can also be optimized by **gradient descent**:

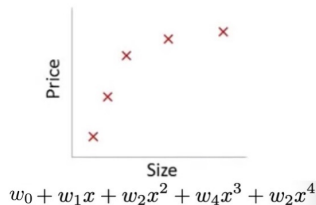
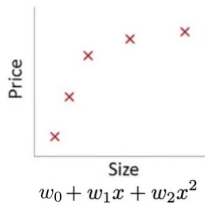
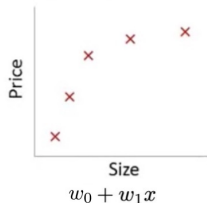
$$\begin{aligned}\mathbf{w}_j &\leftarrow \mathbf{w}_j - \alpha \frac{\partial J(\mathbf{W})}{\partial \mathbf{w}_j}, \\ \text{hint: } \frac{\partial J(\mathbf{W})}{\partial \mathbf{w}_j} &= -\frac{1}{m} \sum_i^m \left[\frac{\mathbb{I}(y_i = j)}{f_{\mathbf{w}_j}(\mathbf{x}_i)} \cdot \frac{\nabla f_{\mathbf{w}_j}(\mathbf{x}_i)}{\nabla \mathbf{w}_j} \right. \\ &\quad \left. + \sum_{c \neq j}^C \frac{\mathbb{I}(y_i = c)}{f_{\mathbf{w}_c}(\mathbf{x}_i)} \cdot \frac{\nabla f_{\mathbf{w}_c}(\mathbf{x}_i)}{\nabla \mathbf{w}_j} \right] \\ \implies \frac{\partial J(\mathbf{W})}{\partial \mathbf{w}_j} &= \frac{1}{m} \sum_i^m (f_{\mathbf{w}_j}(\mathbf{x}_i) - \mathbb{I}(y_i = j)) \mathbf{x}_i\end{aligned}\tag{14}$$

Note: $\{\mathbf{w}_c\}_{c=1}^C$ should be updated **in parallel**, rather than **sequentially**.

- 1 Review of last week
- 2 Classification and representation
- 3 Logistic regression
- 4 Regularized logistic regression**
- 5 Probabilistic perspective of logistic regression
- 6 Summary: linear regression vs. logistic regression

Overfitting in linear regression

Example: Linear regression (housing prices)



Overfitting in linear regression

Addressing overfitting:

x_1 = size of house

x_2 = no. of bedrooms

x_3 = no. of floors

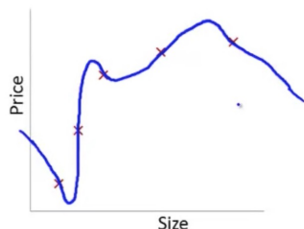
x_4 = age of house

x_5 = average income in neighborhood

x_6 = kitchen size

$\vdots$

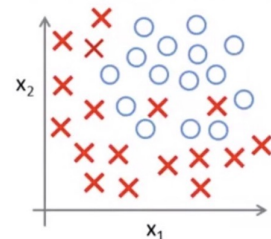
x_{100}



Overfitting: If we have too many features, the learned hypothesis may fit the training data very well (low bias), but fail to generalize to new examples.

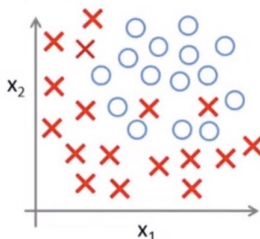
Overfitting in logistic regression

Example: Logistic regression



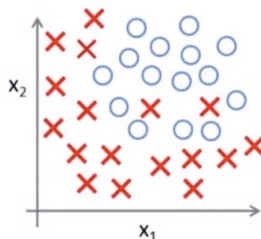
$$f_{\mathbf{w}}(\mathbf{x}) = g(w_0 + w_1x_1 + w_2x_2)$$

Under-fitting



$$f_{\mathbf{w}}(\mathbf{x}) = g(w_0 + w_1x_1 + w_2x_2 + w_3x_1^2 + w_4x_2^2 + w_5x_1x_2)$$

Good-fitting



$$f_{\mathbf{w}}(\mathbf{x}) = g(w_0 + w_1x_1 + w_2x_1^2 + w_3x_1^2x_2 + w_4x_1^2x_2^2 + w_5x_1^2x_2^3 + w_6x_1^3x_2 + \dots)$$

Over-fitting

Addressing Overfitting

Generally, there are two approaches to address the overfitting problem, including:

- Reducing the number of features:
 - Feature selection
 - Dimensionality reduction (introduced in later lectures)
- Regularization:
 - Keep all features, but reduce the magnitude/value of each parameter, such that each feature contributes a bit to predict y

In the following, we will focus on the **regularization-based approach**.

Regularized logistic regression

- The objective function of the regularized logistic regression is formulated as follows

$$\begin{aligned}\bar{J}(\mathbf{w}) &= J(\mathbf{w}) + \frac{\lambda}{2m} \sum_{j=1}^d w_j^2 \\ &= -\frac{1}{m} \sum_i^m [y_i \log(f_{\mathbf{w}}(\mathbf{x}_i)) + (1 - y_i) \log(1 - f_{\mathbf{w}}(\mathbf{x}_i))] + \frac{\lambda}{2m} \sum_{j=1}^d w_j^2.\end{aligned}$$

Note: the bias parameter w_0 is not regularized/penalized.

- The above objective function can also be solved by gradient descent, as follows

$$\begin{aligned}w_0 &\leftarrow w_0 - \frac{\alpha}{m} \sum_{i=1}^m (f_{\mathbf{w}}(\mathbf{x}_i) - y_i) \cdot \mathbf{x}_i(0), \text{ where } \mathbf{x}_i(0) = 1, \forall i \\ w_j &\leftarrow w_j - \frac{\alpha}{m} \left[\sum_{i=1}^m (f_{\mathbf{w}}(\mathbf{x}_i) - y_i) \cdot \mathbf{x}_i(j) + \lambda \cdot w_j \right],\end{aligned}$$

where $\mathbf{x}_i(j)$ denotes the j -th entry of $\mathbf{x}_i$, and $j = 0, \dots, d$.

Regularized logistic regression

Exercise: When using regularized logistic regression, which of these is the best way to monitor whether gradient descent is working correctly?

- Plot $J(\mathbf{w})$ as a function of the number of iterations and make sure it's decreasing
- Plot $J(\mathbf{w}) - \frac{\lambda}{2m} \sum_{j=1}^d w_j^2$ as a function of the number of iterations and make sure it's decreasing
- Plot $J(\mathbf{w}) + \frac{\lambda}{2m} \sum_{j=1}^d w_j^2$ as a function of the number of iterations and make sure it's decreasing
- Plot $\sum_{j=1}^d w_j^2$ as a function of the number of iterations and make sure it's decreasing

- 1 Review of last week
- 2 Classification and representation
- 3 Logistic regression
- 4 Regularized logistic regression
- 5 Probabilistic perspective of logistic regression
- 6 Summary: linear regression vs. logistic regression

Logistic regression: probabilistic modeling

- Behind **logistic regression for binary classification**, we assume that both the feature $\mathbf{x}$ and the label y are random variables, as follows

$$\begin{aligned}\mu(\mathbf{x}; \mathbf{w}) &= \text{Sigmoid}(\mathbf{w}^\top \mathbf{x}), \\ y(\mathbf{x}; \mathbf{w}) &\sim \text{Bernoulli}(\mu(\mathbf{x}; \mathbf{w})).\end{aligned}$$

- Then, we have

$$P(y|\mathbf{x}; \mathbf{w}) = \begin{cases} \mu & \text{if } y = 1, \\ 1 - \mu & \text{if } y = 0. \end{cases}$$

- The **log-likelihood** function of $P(y|\mathbf{x}; \mathbf{w})$ is formulated as

$$\mathcal{L}(\mathbf{w}) = y \log(\mu) + (1 - y) \log(1 - \mu).$$

- Thus, we obtain

$$\max_{\mathbf{w}} \mathcal{L}(\mathbf{w}) \equiv \min_{\mathbf{w}} J(\mathbf{w}).$$

Logistic regression: probabilistic modeling

- Behind [logistic regression](#), we assume that

$$\begin{aligned}\mu(\mathbf{x}; \mathbf{w}) &= \text{Sigmoid}(\mathbf{w}^\top \mathbf{x}), \\ y(\mathbf{x}; \mathbf{w}) &\sim \text{Bernoulli}(\mu(\mathbf{x}; \mathbf{w})).\end{aligned}$$

- [ℓ₂-regularized logistic regression](#): we further assume $\mathbf{w} \sim \mathcal{N}(\mathbf{w}|\mathbf{0}, \sigma^2 \mathbf{I})$ (excluding the bias w_0), then we have

$$\max_{\mathbf{w}} \mathcal{L}(\mathbf{w}) + \log \mathcal{N}(\mathbf{w}|\mathbf{0}, \sigma^2 \mathbf{I}) \equiv \min_{\mathbf{w}} J(\mathbf{w}) + \frac{\lambda}{2m} \sum_{j=1}^d w_j^2.$$

- [ℓ₁-regularized logistic regression](#): if we assume $\mathbf{w} \sim \text{Laplace}(\mathbf{w}|\mathbf{0}, b)$ (excluding the bias w_0), then we have

$$\max_{\mathbf{w}} \mathcal{L}(\mathbf{w}) + \log \text{Laplace}(\mathbf{w}|\mathbf{0}, b) \equiv \min_{\mathbf{w}} J(\mathbf{w}) + \frac{\lambda}{2m} \sum_{j=1}^d |w_j|.$$

- 1 Review of last week
- 2 Classification and representation
- 3 Logistic regression
- 4 Regularized logistic regression
- 5 Probabilistic perspective of logistic regression
- 6 Summary: linear regression vs. logistic regression

Summary: linear regression vs. logistic regression

	Linear regression	Logistic regression
Task	regression	classification
Hypothesis $f_{\mathbf{w}}(\mathbf{x})$	$\mathbf{w}^\top \mathbf{x} \in (-\infty, \infty)$	$g(\mathbf{w}^\top \mathbf{x}) \in [0, 1]$
Objective $J(\mathbf{w})$	$\frac{1}{2m} \sum_i^m (y_i - \mathbf{w}^\top \mathbf{x}_i)^2$	$-\frac{1}{m} \sum_{i=1}^m [y_i \log(f_{\mathbf{w}}(\mathbf{x}_i)) + (1 - y_i) \log(1 - f_{\mathbf{w}}(\mathbf{x}_i))]$
Solution	closed-form or gradient descent	gradient descent

Note that: For each variant of linear/logistic regression, you can derive it from both the deterministic and the probabilistic perspectives.

Own reading: Both linear regression and logistic regression are special cases of **generalized linear models**. If interested, you can find more details in Section 4 of the book “Pattern Recognition and Machine Learning”, Bishop, 2006.